Glossary: AI Terms for Beginners

Front Matter, Chapter 12/12 (position #14) — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/glossary (generation 82)

EN: AI contains ML, which contains DL, which contains LLMs -- each a subset of the last. This is a pure reference chapter (no projects/ folder): Top 30 Terms, then the Architecture Runtime Stack, then 15 topic categories. Use Ctrl+F, not front-to-back reading.
Roman Urdu: AI mein ML hai, ML mein DL, DL mein LLMs -- har ek pichle ka subset hai. Yeh pure reference chapter hai (koi projects/ nahi): pehle Top 30 Terms, phir Architecture Runtime Stack, phir 15 topic categories. Ctrl+F use karo, front-to-back mat parho.

1. Architecture: The Runtime Stack

- One-sentence taxonomy: **Agent Factory** (process) -> **AI-Native Company** (output) -> **AI Workers** (workforce) -> **Two-Layer Model** (Edge + AI Workforce layers) -> runtime engine -> triggers.
- **Delegate (OpenClaw)** = chief-of-staff, one per executive (Edge Layer). **Manager (Paperclip)** = COO, one-to-many over the workforce (AI Workforce Layer). Both are permanent staff, NOT part of the hire/fire workforce.
- **Meta-Layer** = hiring as a callable capability -- Paperclip detects a gap, calls a hiring API inside its authority envelope, a new Worker is manufactured without waking a human.
- **Harness** = the control plane (agent loop, tool dispatch, approvals, tracing, context, recovery). Model is the 'brain', harness is the 'body + workplace + SOP'. Credentials live in the harness; model-generated code runs in the **sandbox** -- split for security + portability.

2. Agent Factory Book-Specific Terms (Category 1)

- **Agent Factory**, **Industrialized Stack** (Intent -> Production Engine -> Outcome), **Six Invariants**, **Invariant vs Reference Implementation** (structural requirement that never changes vs. the concrete product realizing it today), **Digital Twin**, **Digital FTE**, **Spec/SDD/TDG**, **10-80-10 Rule**, **AGENTS.md/CLAUDE.md**.

3. The Other 14 Categories (Overview)

- Core AI/ML, LLM Basics, Knowledge/Retrieval/Context (RAG, vector DBs, MCP), Agentic AI Concepts (chatbot vs agent, multi-agent), Programming/Software Terms, Data/Database Terms.
- Cloud/Deployment (Docker, CI/CD), Realtime/Voice Agent Terms, Security/Safety/Enterprise Terms, Agentic Commerce & Payments (ACP/AP2/x402/MPP), Monitoring/Quality/LLMOps.
- Protocols & Standards (AAIF, A2A, OpenAPI), Business/Product/Strategy Terms, Tools & Products Referenced (AWS/GCP/Azure/Cloudflare/VS Code/Markdown).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
AI	Making computers do things that normally require human intelligence.	Computers se woh kaam karana jo normally human intelligence chahta hai.
LLM	A giant AI system trained on billions of pages of text.	Bohot zyada text par trained bara AI system (Claude, GPT, Gemini).
Agent (AI Agent)	AI that takes action, makes plans, gets things done on its own.	AI jo action leta, plan banata, khud kaam poora karta hai.
Agentic AI	The category focused on agents that plan, reason, and act autonomously.	Agents banane wali category jo plan/reason/act khud karte hain.
Digital FTE	An AI employee doing continuous full-time work, 24/7, cheaply.	AI employee jo 24/7 continuous kaam karta hai, sasta.
Agent Factory	The spec-driven, human-supervised process that manufactures AI Workers.	Spec-driven, human-supervised process jo AI Workers manufacture karta hai.
Prompt	The instruction or question you type into an AI model.	AI model mein type ki gayi instruction ya sawal.
Context Window	How much text the AI can read and think about at once.	AI ek waqt mein kitna text parh/soch sakta hai.
Token	The basic text unit an LLM reads, ~3/4 of a word.	LLM ka basic text unit, ek lafz ka taqreeban 3/4.
Hallucination	When AI confidently generates something untrue.	Jab AI confidently koi jhoothi baat generate kare.
Spec (Specification)	A detailed blueprint of what you want built.	Jo banana hai uska detailed blueprint.
Spec-Driven Development (SDD)	Write the blueprint first, let AI generate code from it.	Pehle blueprint likho, phir AI us se code generate kare.
MCP	The universal protocol letting any agent plug into any tool.	Universal protocol jo kisi bhi agent ko kisi bhi tool se jorta hai.
Context Engineering	Designing the full info environment an agent receives.	Agent ko milne wala poora info environment design karna.
Tool Use	An agent's ability to use external tools instead of just answering from memory.	Agent ka external tools use karne ka ability, sirf memory se nahi.
Guardrails	Safety constraints preventing an agent from doing things it shouldn't.	Safety limits jo agent ko galat kaam se rokte hain.
RAG	Giving AI external documents so it answers from facts, not memory.	AI ko external documents dena taake wo facts se jawab de, memory se nahi.
10-80-10 Rule	Human sets direction (10%), AI executes (80%), human verifies (10%).	Insaan direction deta (10%), AI execute karta (80%), insaan verify karta (10%).
Orchestration	Coordinating multiple agents to work together.	Multiple agents ko mil kar kaam karwana.
Stateless	AI forgets everything between conversations.	AI conversations ke beech sab bhool jata hai.
CI/CD	Automatically testing and deploying code on every change.	Har change par code automatically test/deploy hona.

Quick Self-Check (from the chapter's own 15-Q quiz)

Question	Answer
What's the nesting relationship AI/ML/DL/LLM?	AI contains ML, ML contains DL, DL contains LLMs -- each narrower and more specific.
Difference between a chatbot and an agent?	A chatbot answers questions; an agent plans, researches, executes, and reports back, making decisions along the way.
What's the difference between Delegate (OpenClaw) and Manager (Paperclip)?	Delegate = one-per-executive chief-of-staff (Edge Layer); Manager = one-to-many COO over the workforce (AI Workforce Layer).
What's an Invariant vs a Reference Implementation?	Invariant = the structural requirement that must always hold; reference implementation = today's concrete product meeting it (the implementation can change, the invariant can't).
Why does the harness hold credentials while code runs in the sandbox?	Security + portability -- the sandbox vendor can be swapped without rewriting the agent.
What problem does MCP solve?	A universal protocol so any agent can plug into any tool -- like USB replacing one-charger-per-phone.
Why is RAG described as an 'open-book exam'?	AI looks up facts in your documents before answering instead of guessing from (possibly wrong) memory -- much more accurate.
What does 'stateless' mean for a chat app's apparent memory?	The AI forgets everything between chats; apps fake memory by re-sending the whole conversation each time.

Where This Leads / Sources

Leads to: every other chapter in this repo (Top 30 Terms appear on nearly every page). Companion reference: docs/cheatsheets/.

Source: docs/glossary/ (this repo, 18 files, 15 categories), content originally via Zia Tutor AI connector.